

Month 2: Purple Team Batch 01

Lateral Movement & Defensive Engineering

Week 5: Network Detection & Defense Engineering

Instructor: **Muhammad Saad**

IDS - Intrusion Detection System

An **Intrusion Detection System (IDS)** is a hardware or software solution that continuously observes a computer system or network to detect malicious activities such as hacking attempts, malware, or unauthorized access.

Types of IDS

Network-Based IDS (NIDS)

- Monitors traffic across an entire network
- Installed at strategic points (like routers or gateways)
- Detects attacks like DDoS or port scanning

Host-Based IDS (HIDS)

- Runs on individual devices (computers/servers)
- Monitors system files, logs, and processes
- Detects insider threats or file changes

Detection Methods

Signature-Based Detection

- Compares traffic with known attack patterns
- Fast but cannot detect new (unknown) attacks

Anomaly-Based Detection

- Learns normal behavior and flags unusual activity
- Can detect new threats but may give false alarms

Hybrid Detection

- Combines both methods for better accuracy

IPS - Intrusion Prevention System

An **Intrusion Prevention System (IPS)** is a hardware or software solution that monitors network or system activities and **automatically takes action** (such as blocking traffic or terminating connections) when it detects malicious behavior.

Types of IDS

Network-Based IPS (NIPS)

- Monitors entire network traffic
- Placed inline within the network
- Blocks harmful packets before they reach targets

Host-Based IPS (HIPS)

- Installed on individual systems
- Protects against internal threats and unauthorized changes

Wireless IPS (WIPS)

- Monitors wireless networks
- Detects rogue access points and Wi-Fi attacks

Network Behavior Analysis IPS (NBA)

- Detects abnormal traffic patterns
- Useful for identifying DDoS attacks

Detection Methods

Signature-Based Detection

- Matches known attack patterns
- Effective against known threats

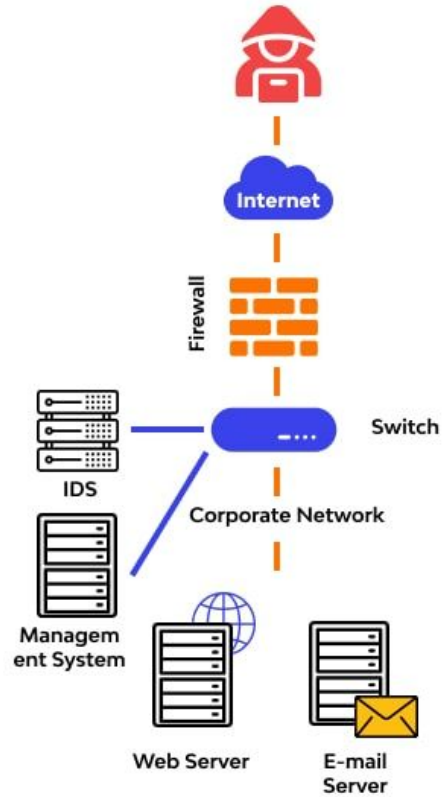
Anomaly-Based Detection

- Detects unusual behavior
- Can identify new or unknown attacks

Policy-Based Detection

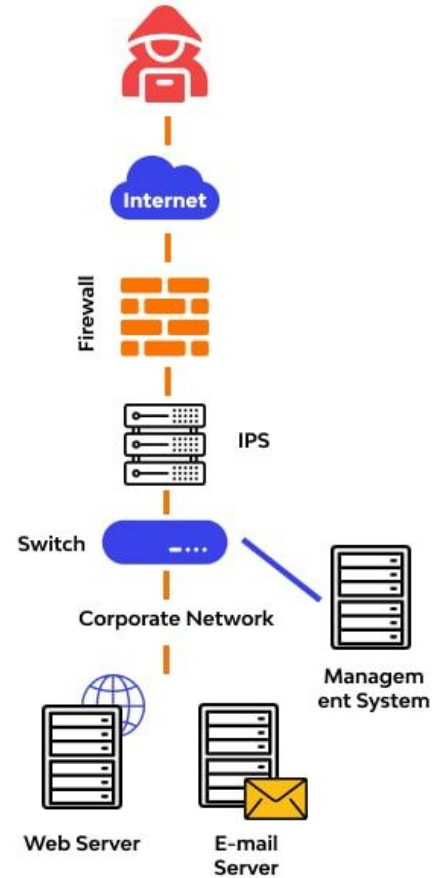
- Enforces predefined security rules
- Blocks actions that violate policies

Intrusion Detection System (IDS)



VS

Intrusion Prevention System (IPS)



IDS (Out-of-Band Mode)

An **Intrusion Detection System (IDS)** works in **out-of-band mode**, meaning it is **not directly in the path of network traffic**. It only monitors a copy of the traffic.

How it works

- Connected to a network via a **SPAN/mirror port or network tap**
- Receives a **copy of data packets**
- Analyzes traffic and **generates alerts only**

Key Features

- Does **not affect network performance**
- Cannot block or stop attacks
- Passive monitoring system

IPS (In-Line Mode)

An **Intrusion Prevention System (IPS)** works in **in-line mode**, meaning it is placed **directly in the path of network traffic**.

How it works

- Installed between source and destination (e.g., between firewall and network)
- All traffic must pass **through the IPS**
- Can **inspect and take action immediately**

Key Features

- Can **block, drop, or modify malicious traffic**
- Provides **real-time protection**
- May introduce slight delay in traffic

Best of Luck for the Week 5 Tasks

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